

GEOGRAPHY

PHYSICAL GEOGRAPHY



GEOMORPHOLOGY

MODULE - 38

COASTAL LANDFORMS

- **SUBMERGENT COASTS**
- **EMERGENT COASTS**

GEOMORPHOLOGY

PLAN OF ACTION

WHAT WE HAVE COVERED

- EARTH'S INTERIOR
- SOURCES OF HEAT IN EARTH'S INTERIOR
- CONTINENTAL DRIFT THEORY
- PLATE TECTONICS THEORY
- CONVERGENT, DIVERGENT & TRANSFORM PLATE BOUNDARIES

- VOLCANOES
- EARTHQUAKES
- GEOMORPHIC PROCESSES
- LANDFORMS & THEIR EVOLUTION
 1. FLUVIAL LANDFORMS
 2. GLACIAL LANDFORMS
 3. ARID LANDFORMS
 4. KARST TOPOGRAPHY
 5. COASTAL LANDFORMS

THE MECHANISM OF WAVE EROSION

WAVES AND CURRENTS



- 1** Coastal processes are the most dynamic and hence most destructive.
- 2** Some of the changes along the coastal take place very fast.
- 3** At one place, there can be erosion in one season and deposition in another.
- 4** Most of the changes along the coasts are accomplished by waves.

THE MECHANISM OF WAVE EROSION

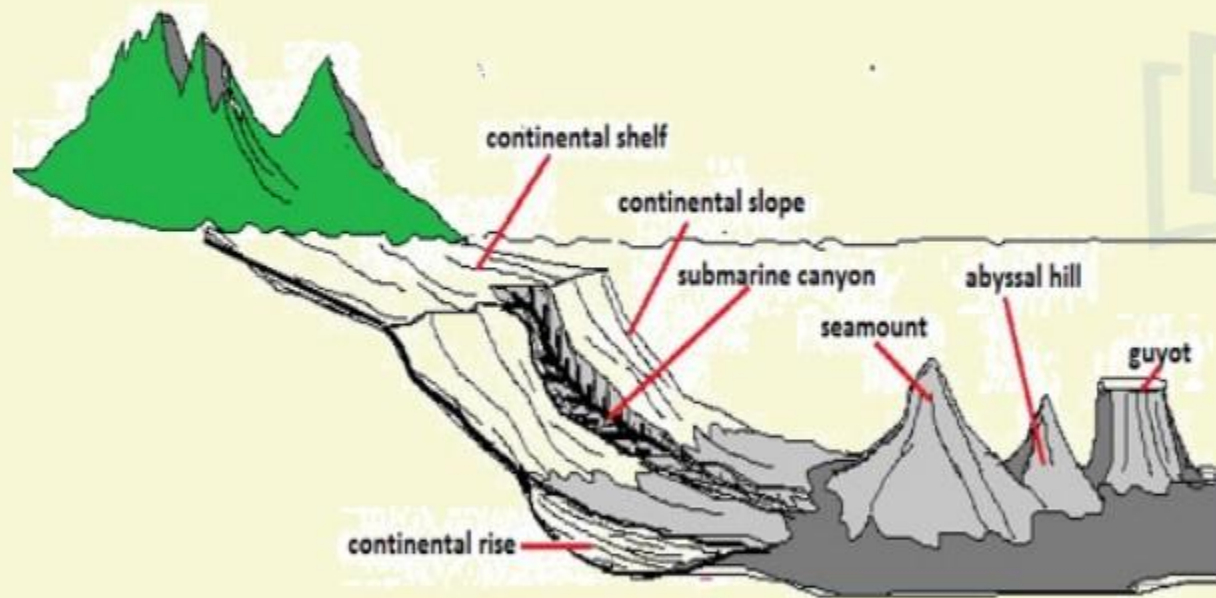
WAVES AND CURRENTS



- 5** When waves break, the water is thrown with greater force onto the shore, and simultaneously, there is a **great churning of sediments** on the sea bottom.
- 6** Constant impact of breaking waves drastically affects the coasts.
- 7** **Storm waves and tsunami waves** can cause far-reaching changes in a short period of time than normal breaking waves.

THE MECHANISM OF WAVE EROSION

WAVES AND CURRENTS



8 Other than the action of waves, the coastal landforms depend upon

- i. Configuration of land and sea floor
- ii. Whether the coast is advancing (emerging) seaward or retreating (submerging) landward.

9 Assuming sea level to be constant, two types of coasts are considered to explain the concept of evolution of coastal landforms

- i. High, rocky coasts (submerged coasts)
- ii. Low, smooth and gently sloping sedimentary coasts (emerged coasts)

COASTAL LANDFORMS

SUBMERGENT COASTS vs EMERGENT COASTS



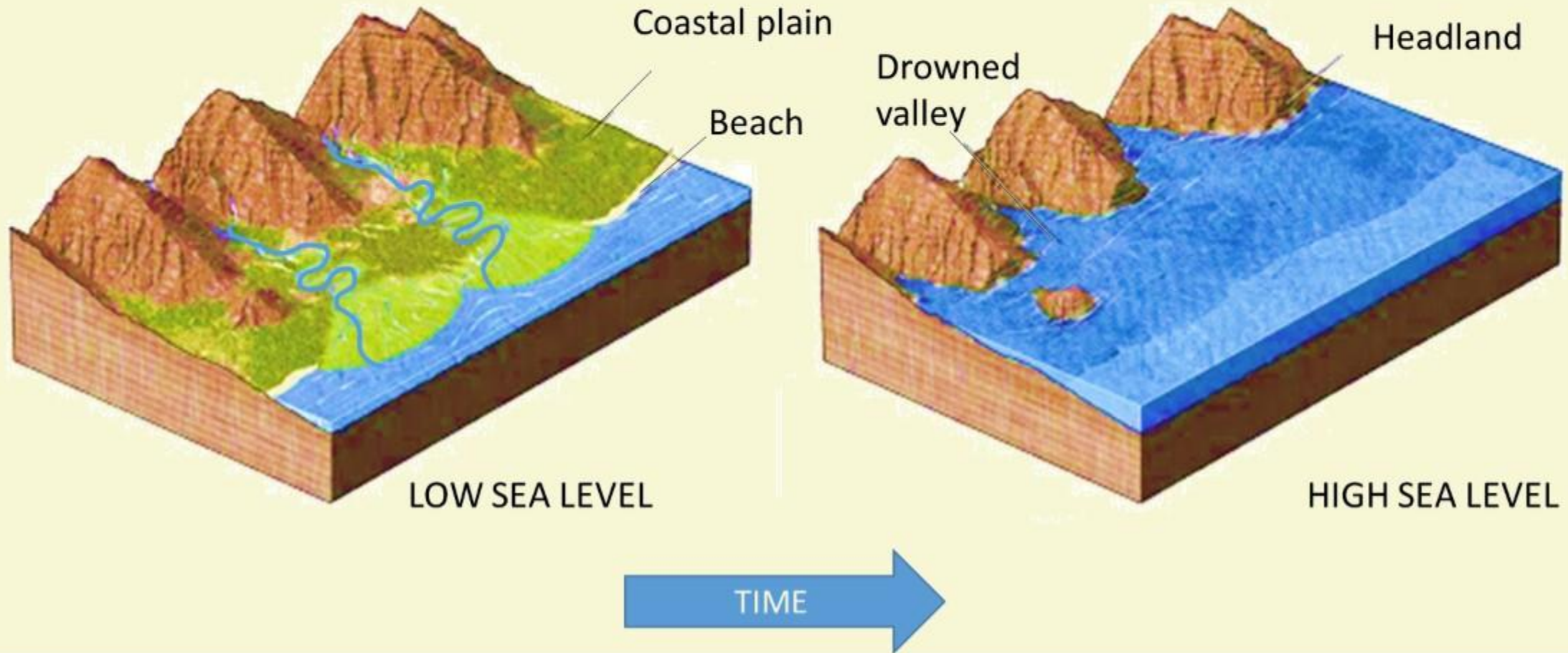
EMERGENT COASTS



SUBMERGENT COASTS

1 HIGH ROCKY COASTS/ SUBMERGENT COASTS

SUBMERGENT COASTS: Retreating coasts. Rivers drain valleys and deposit sediments on a coastal plain. As a submergent coast forms, sea-level rises and flood the valleys and waves start eroding the headlands.



COASTAL LANDFORMS

ACTION OF WAVES ON COASTS

1 HIGH ROCKY COASTS/ SUBMERGENT COASTS



- 1 Along the high rocky coasts, **the rivers appear to have been drowned** with highly irregular coastline.
- 2 The coastline appears **highly indented** with extension of water into the land. (fjord formation takes place)
- 3 The hill sides drop off sharply into the water.
- 4 Shores do not show any depositional landforms initially. **Erosional features dominate.**

COASTAL LANDFORMS

ACTION OF WAVES ON COASTS

1 HIGH ROCKY COASTS/ SUBMERGENT COASTS



5 Along high rocky coasts, waves break with great force against the land shaping the hill sides into cliffs.

6 With constant pounding by waves, the cliffs recede leaving a wave-cut-platform in front of the sea cliff. Waves gradually minimise the irregularities along the shore.

7 The materials which fall off, and removed from the sea cliffs, gradually break into smaller fragments and roll to roundness, will get deposited in the offshore.

COASTAL LANDFORMS

ACTION OF WAVES ON COASTS

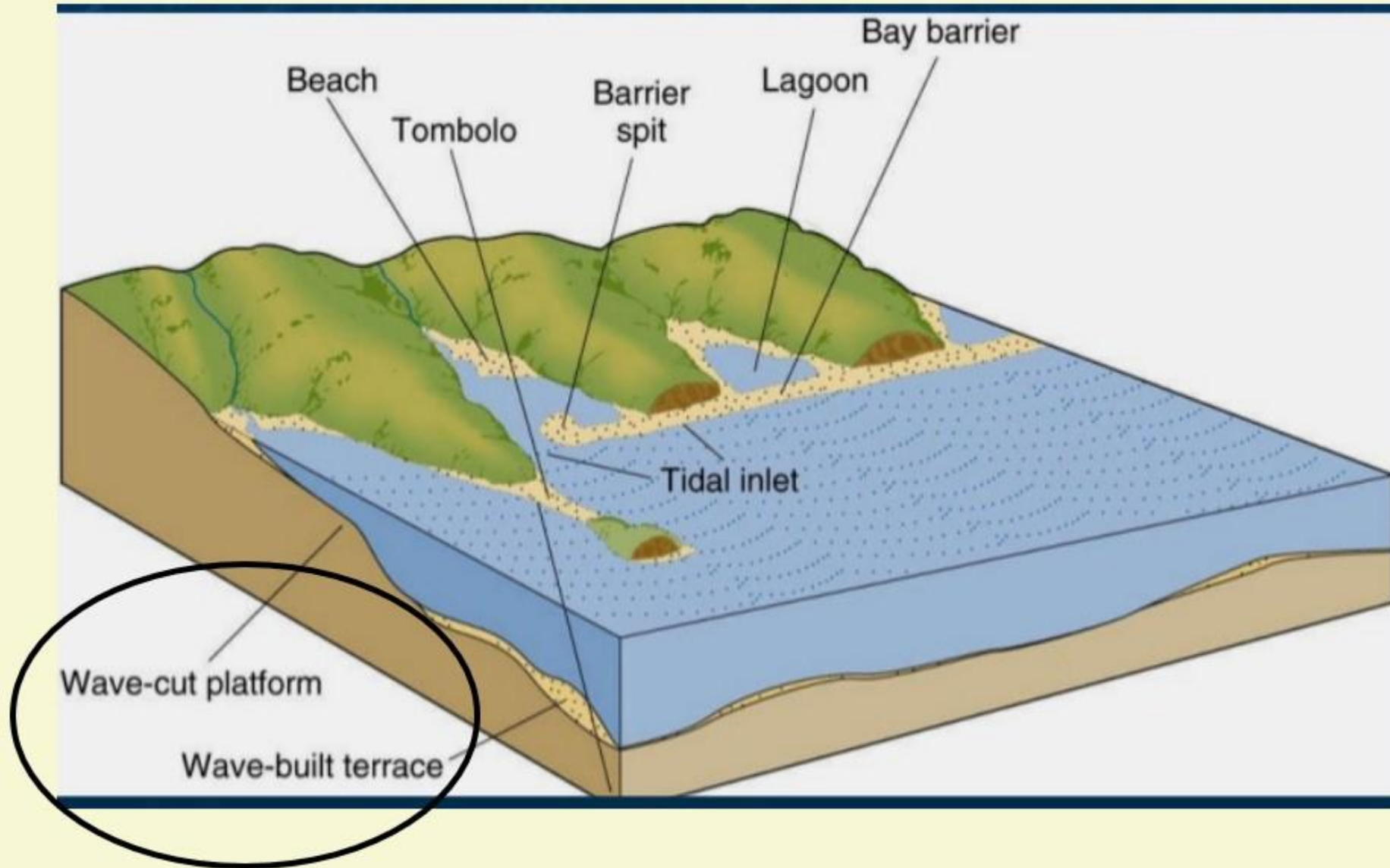
1 HIGH ROCKY COASTS/ SUBMERGENT COASTS



8 After a considerable period of cliff development and retreat when coast line turns somewhat smooth, with the addition of some more material to this deposit in the offshore, a wave-built terrace would develop in front of wave-cut terrace.

9 As the erosion along the coast takes place, a good supply material becomes available to long shore currents and waves to deposit as beaches along the shore and as bars (long ridges of sand and/or shingle parallel to the coast) in the near shore zone.

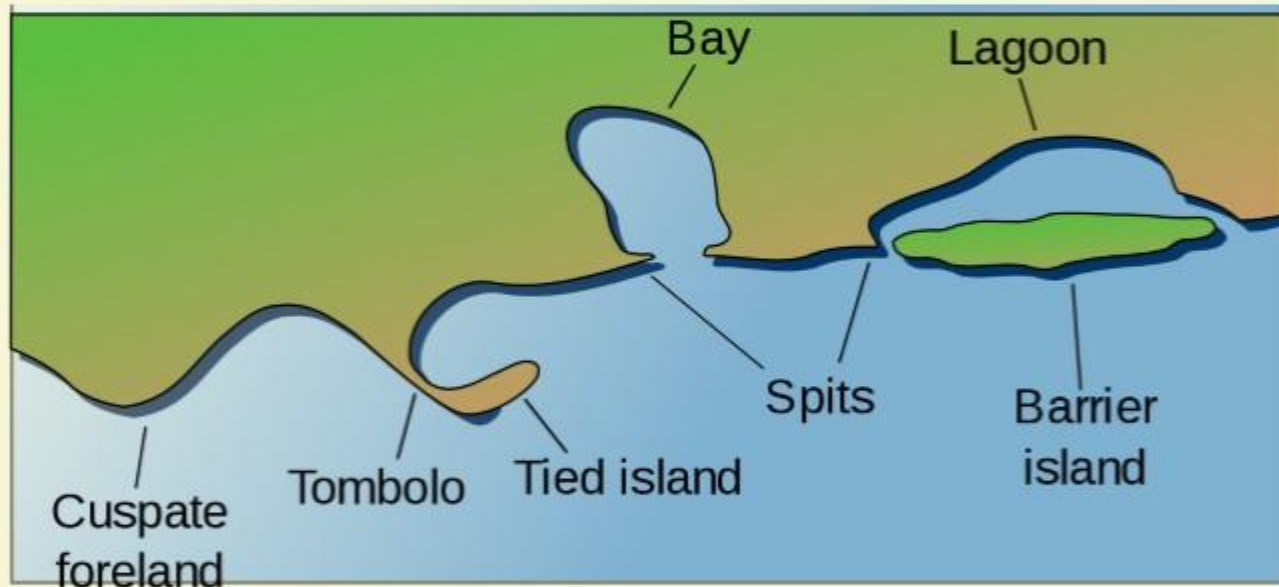
WAVE-CUT vs WAVE-BUILT PLATFORM



COASTAL LANDFORMS

ACTION OF WAVES ON COASTS

1 HIGH ROCKY COASTS/ SUBMERGENT COASTS



11 Bars are submerged features initially and when bars show up above water, they are called *barriers bars*.

12 Barrier bar which get keyed up to the headland of a bay is called a *spit*.

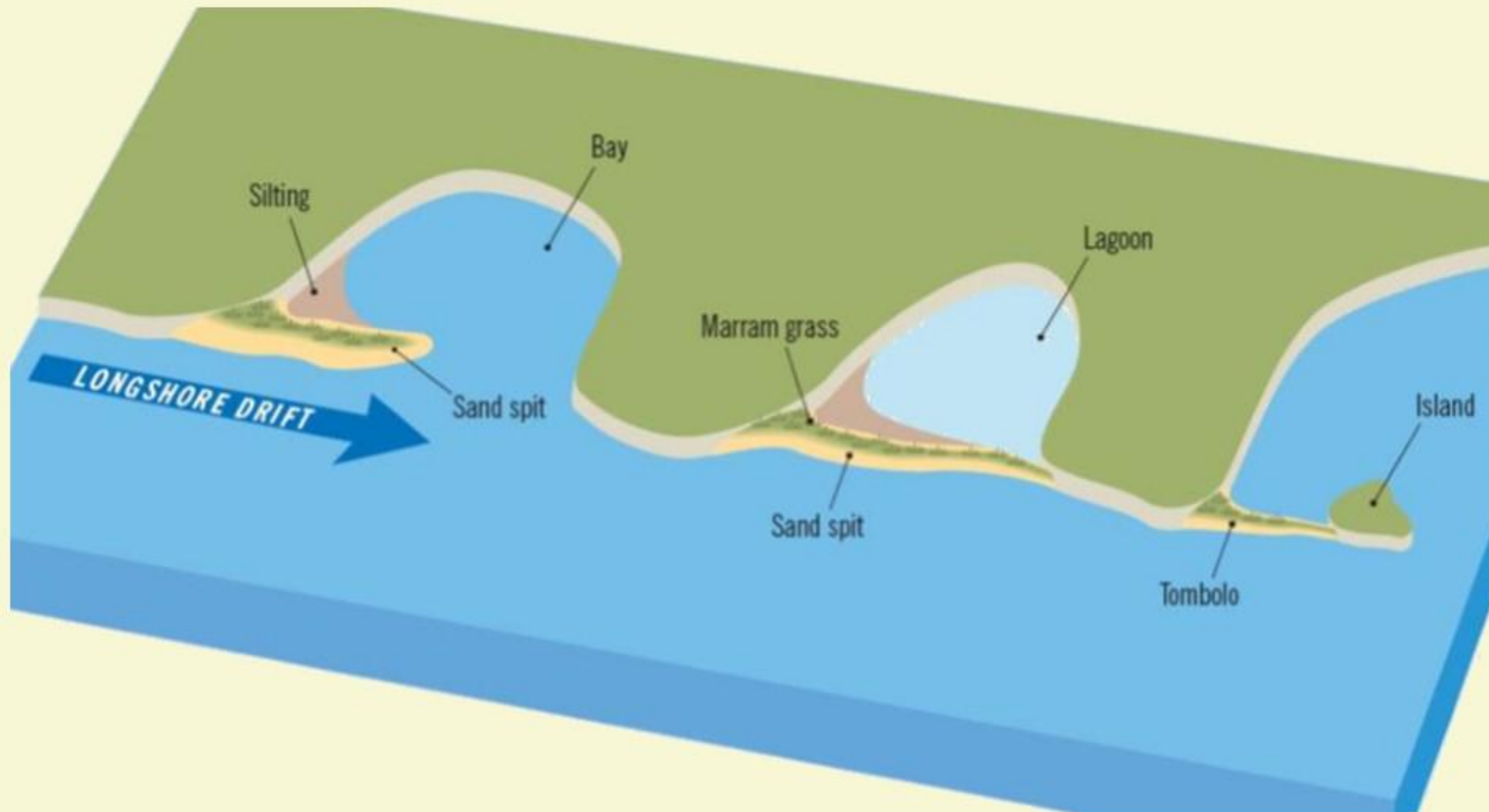
13 When barrier bars and spits form at the mouth of a bay and block it, a *lagoon forms*.

14 The lagoons would gradually get filled up by sediments from the land giving rise to a coastal plain.

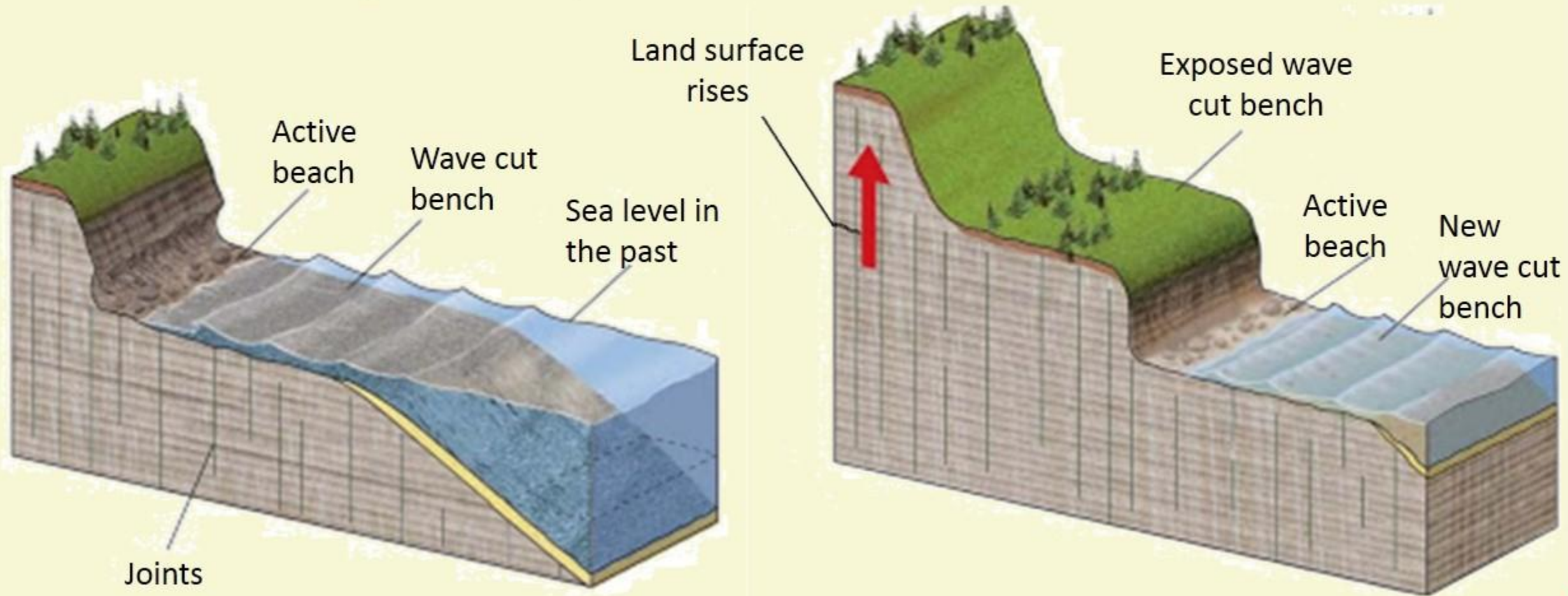
COASTAL LANDFORMS

ACTION OF WAVES ON COASTS

1 HIGH ROCKY COASTS/ SUBMERGENT COASTS



EMERGENT COASTS: Wave erosion produces a wave-cut bench along an emergent coast. As the land rises, the bench becomes a terrace and a new wave-cut bench forms



COASTAL LANDFORMS

ACTION OF WAVES ON COASTS

2 LOW ROCKY COASTS/ EMERGENT COASTS

1 Along low rocky coasts, the rivers appear to extend their length by building coastal plains and deltas.



2 The coastline appears smooth with occasional incursions of water in the form of lagoons and tidal creeks.

3 The land slopes gently into the water. Marshes and swamps may abound along the coasts. Depositional features dominate.

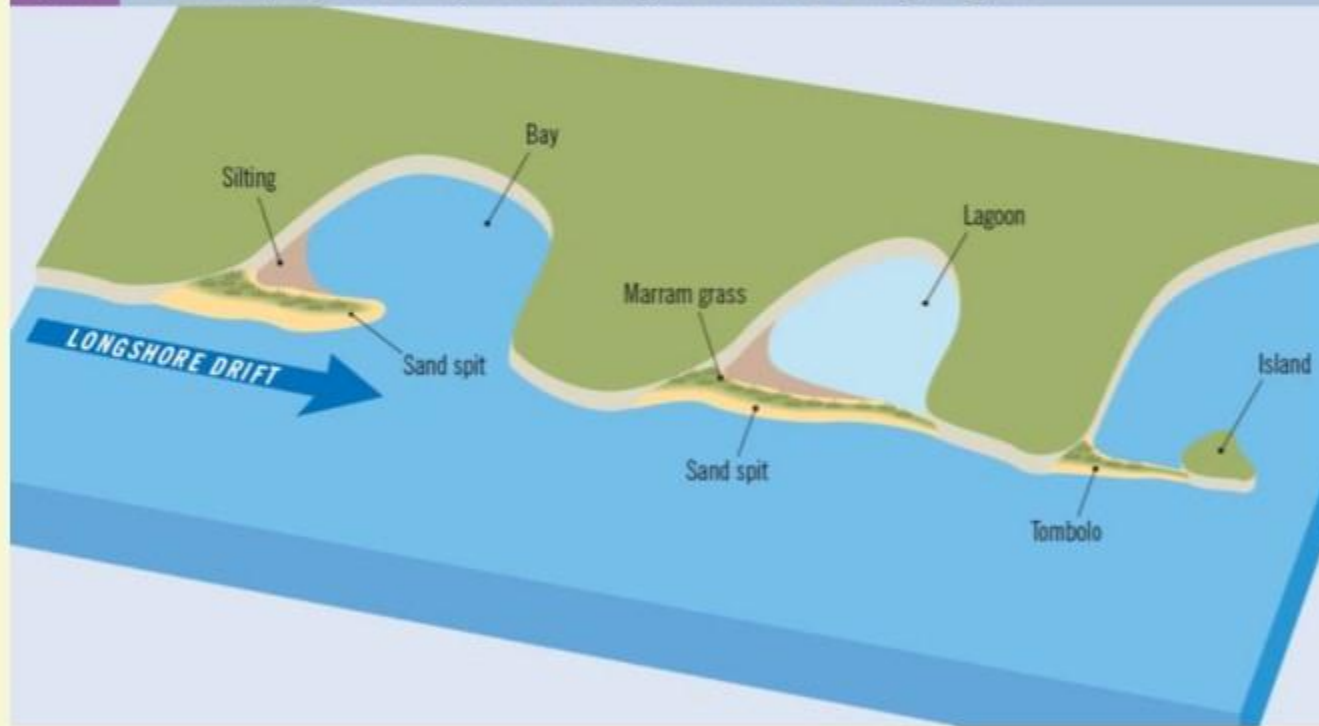
4 When water breaks over a gently sloping sedimentary coast, the bottom sediments get churned and move readily building bars, barrier bars, spits and lagoons.

COASTAL LANDFORMS

ACTION OF WAVES ON COASTS

2 LOW ROCKY COASTS/ EMERGENT COASTS

Fig 2.33 If a sand spit grows, it may block off a bay from the sea, forming a lagoon.



- 5 Lagoons would eventually turn into a swamp which would subsequently turn into a coastal plain.
- 6 The maintenance of these depositional features depends upon the **steady supply of materials**.
- 7 Storm and tsunami waves cause drastic change irrespective of supply of sediments.
- 8 Large rivers which bring lots of sediments build deltas along low sedimentary coasts.

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For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

OUR TEAM

G. V. Rao

K. Srikanth

D. Sunil Kumar

L. V. Krishna

Y. Deepthi

Amrita Naidu

M. Binukrishna

G. Ravi Babu

D. Joji

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